

6 Results III: Transferability Hypothesis

6.1 Block 3 objective and evidence boundary

Blocks 1 and 2 established the source-case recipe on BOPTEST `bestest_air`: a v3 rollout surrogate supplies smooth control-oriented dynamics, a calibrated v3.5 RC-NeuralODE supplies a frozen physical disagreement censor, and the thermostatic PPO policy trained on the resulting hybrid backend is the strongest verified controller on the targeted windows. Block 3 asks a narrower, more falsifiable question: does this fixed source-case recipe transfer to related BOPTEST hydronic-family testcases?

The claim boundary is deliberately limited. Block 3 does not claim universal building generalization, cross-climate generalization, or transfer to arbitrary HVAC topologies. It evaluates transfer to three pre-selected single-zone hydronic-family cases under documented actuator adapters and pre-registered recalibration regimes, with controller fine-tuning on the target testcase explicitly excluded. That exclusion is methodologically central: it separates *controller* transfer from *surrogate-recalibration* transfer, so the evidence is reported component-wise.

Roadmap boundary and executed path. Block 3 (`roadmap.md` Sections 14–15) was opened only after the Block 1/2 article state was committed, and runs strictly as:

1. pre-register the testcase/regime/hypothesis manifest before any non-`bestest_air` run (§14);
2. for each testcase, smoke-test the actuator adapter, run the PI baseline, and run `mode=none` frozen-controller transfer (§15.1–15.2);
3. collect target telemetry and run partial (Stage C) and full (Stage A/B/C) surrogate recalibration (§15.2);
4. aggregate the transfer matrix and close the pre-registered hypotheses (§15.3–15.5).

Artifact provenance and rebuild commands for every table and figure are catalogued in roadmap Section 15.

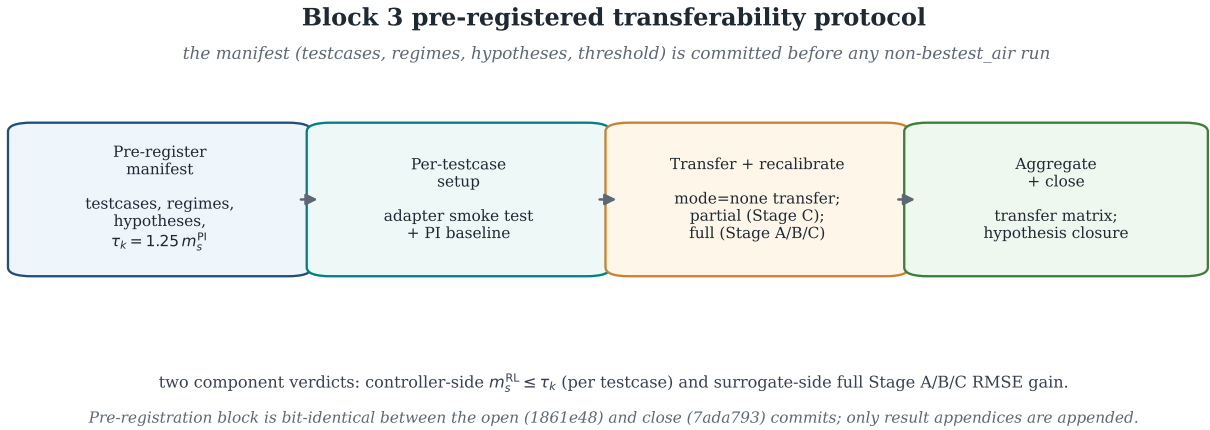


Figure 1: Block 3 pre-registered transferability protocol. Controller-side transfer is tested against a testcase-specific PI threshold; surrogate-side transfer is tested by full Stage A/B/C recalibration on target telemetry.

Table 1: Nomenclature for Block 3.

Symbol	Unit	Meaning
$m_{s,RL}$	–	live BOPTEST maintenance score of the frozen RL controller (lower better)
$m_{s,PI}$	–	maintenance score of the testcase built-in PI controller
τ_k	–	pre-registered pass threshold $1.25 m_{s,PI,k}$ for testcase k
ΔE_k	%	energy change of the RL controller vs PI on testcase k
G_k^{RMSE}	%	rollout-RMSE improvement after full Stage A/B/C recalibration
$\rho_{C,k}$	–	re-identified C_{zon} ratio vs the bestest_air baseline
$\mathcal{A}_k$	–	pre-registered actuator adapter for testcase k

6.2 Pre-registration and audit anchors

Block 3 is pre-registered through `configs/block3_testcase_manifest.yaml`. The initial manifest commit (1861e48) was made before any non-**bestest_air** BOPTEST run; the pre-registration block is bit-identical between that commit and the close commit, with only result appendices added. The audit anchors are: 1861e48 (pre-registration manifest), 2f9d596 (record pre-registration SHA), eb7091e / 46fbaa9 / 645626e (the three actuator adapters and the stretch-testcase predictions), 7ada793 (close SHA), and cb7025f (component-level interpretation). Because the hypothesis definitions were frozen before the runs, every number below was predictable but not predicted.

6.3 Testcases, actuator adapters, and recalibration regimes

The three target testcases span an increasing difficulty ladder relative to **bestest_air**; all are single-zone or single-zone-like, heating-capable, and structurally non-identical to the source case.

Table 2: Pre-registered target testcases and actuator adapters (manifest `testcase_candidates`).

Role	Testcase	Structural difference vs bestest_air	Adapter
Primary	bestest_hydronic_heat_pump	hydronic loop driven by a heat pump	hydronic_t_supply_v1
Secondary	bestest_hydronic	boiler/radiator hydronic distribution	hydronic_direct_supply_v1
Stretch	singlezone_commercial_hydronic	order-of-magnitude larger commercial zone + hydronic valve	commercial_hydronic_valve_v1

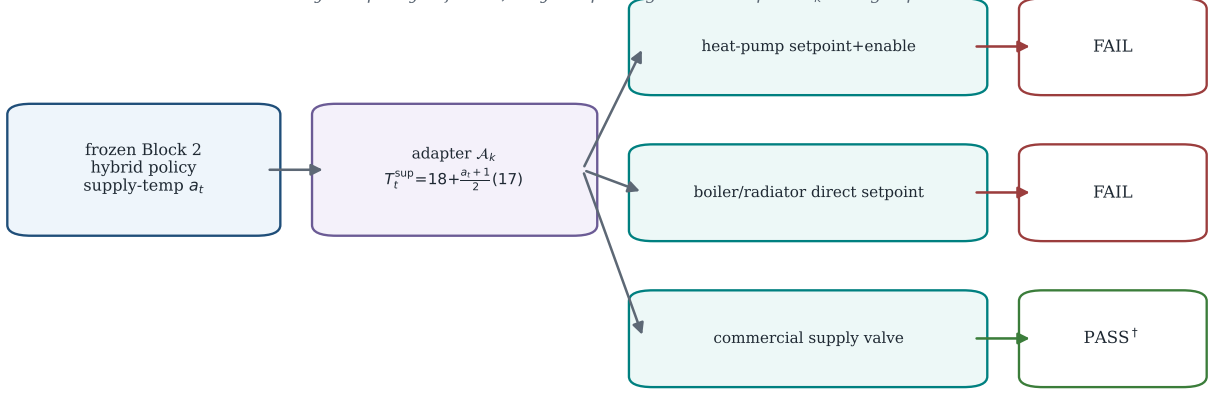
The adapter maps the frozen source policy output to the target actuator interface:

$$u_t^{\text{target}} = \mathcal{A}_k(T_t^{\text{sup}}, y_t), \quad T_t^{\text{sup}} = 18 + \frac{a_t+1}{2}(35 - 18), \quad (1)$$

where $\mathcal{A}_k$ is the pre-registered adapter for testcase k , $a_t \in [-1, 1]$ is the frozen policy output, and y_t are available BOPTEST measurements used by rule-based adapter logic. Adapter smoke tests verify that low/high overrides produce directionally valid heating before any yearly run. Figure 2 shows the adapter-mediated transfer: one frozen controller drives three different actuator interfaces through the per-testcase adapters, with per-testcase controller verdicts. The recalibration regimes (Table 4) keep the controller frozen and vary only how much of the surrogate adapts.

Adapter-mediated transfer: one frozen controller, three actuator interfaces

the Block 2 hybrid policy is frozen; only the pre-registered adapter $\mathcal{A}_k$ changes per testcase



controller-side verdict is per-testcase ($1.25 \times \text{PI}$ threshold); the surrogate side (full Stage A/B/C) PASSES on all three.

Figure 2: Adapter-mediated transfer schematic. The frozen Block 2 hybrid policy is routed through a pre-registered actuator adapter $\mathcal{A}_k$ to each testcase’s actuator interface; the controller-side verdict (against the $1.25 \times \text{PI}$ threshold) is per-testcase, while full Stage A/B/C surrogate recalibration passes on all three.

Concretely, each adapter first maps the policy command to a normalized heat-intensity $h = \text{clip}((T_t^{\text{sup}} - 18)/17, 0, 1)$ and then drives the testcase’s pre-registered BOPTEST override inputs (Table 3). The mappings were finalized and committed before any control run (audit anchors `eb7091e` / `46fbaa9` / `645626e`), so the actuator interface cannot be tuned post hoc.

Table 3: Per-testcase actuator-adapter mapping (verified from `configs/block3_actuator_mapping_*.yaml`). All adapters share $h = \text{clip}((T_t^{\text{sup}} - 18)/17, 0, 1)$; setpoint overrides are in kelvin.

Testcase	Primary override	Override formula	Auxiliary overrides
heat_pump	oveTSet_u (K)	$288.15 + 9h$ (zone setpoint $15\text{--}24^\circ\text{C}$)	oveHeaPumY_u, ovePum_u, oveFan_u from h
hydronic	oveTSetSup_u (K)	$T_t^{\text{sup}} + 273.15$ (direct supply)	ovePum_u from h ; fixed comfort-band context
commercial	dh_oveTSetSupSetHea_u (K)	$T_t^{\text{sup}} + 273.15$ (district supply)	heating valves + pump from h ; fixed zone/air context

Table 4: Pre-registered recalibration regimes. The controller is frozen in every regime.

Regime	Surrogate action	Controller action
none	frozen <code>bestest_air</code> surrogate recipe	frozen Block 2 hybrid controller
partial	Stage C residual-head recalibration only	frozen Block 2 hybrid controller
full	Stage A + Stage B + Stage C on target telemetry	frozen Block 2 hybrid controller

6.4 Testcase architecture and physical processes

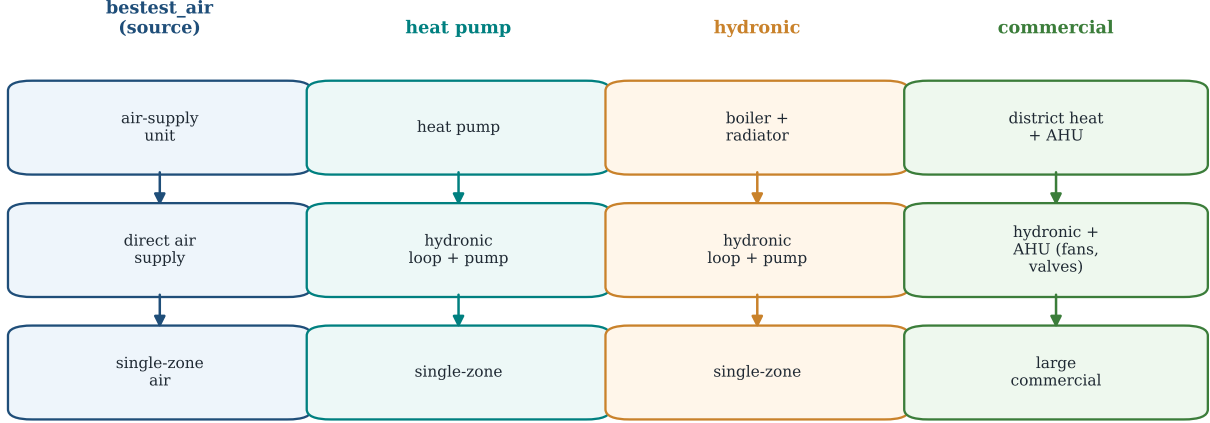
The three targets differ in heat source, distribution path, and actuator set, but share the same single-zone energy balance (Figure 3). For a hydronic zone the governing first-order balance is

$$C_{\text{zon}} \frac{dT_{\text{zon}}}{dt} = \dot{Q}_{\text{hyd}}(u_t) + \frac{T_{\text{amb}} - T_{\text{zon}}}{R_{\text{env}}} + \dot{Q}_{\text{gain}}, \quad (2)$$

where $\dot{Q}_{\text{hyd}}$ is the delivered hydronic heat (whose actuation path — heat pump, boiler/radiator, or district valve — changes per testcase), R_{env} the envelope resistance, and $\dot{Q}_{\text{gain}}$ the internal/solar gains.

Transfer changes the *delivery path* $\dot{Q}_{\text{hyd}}$ and the actuator interface, but C_{zon} is the shared physical invariant that Stage B re-identifies — which is why its re-identified value transfers (Section 6.10).

HVAC topology: source → distribution → zone across the source case and three hydronic targets



Transfer changes the heat source, the distribution path, and the actuator set; the zone energy balance and C_{zon} are the shared physics.

Figure 3: Comparative HVAC topology of the source case and the three hydronic targets (source → distribution → zone). Transfer changes the heat source, distribution path, and actuator set, while the zone energy balance (2) and C_{zon} are shared.

Table 5 summarizes the architecture and the identified physics. The equivalent thermal mass $C_{\text{zon}}/c_{p,\text{air}}$ (with $c_{p,\text{air}} = 1005 \text{ J kg}^{-1} \text{ K}^{-1}$) is roughly $2\times$ the **bestest_air** value for every hydronic case, consistent with the added water-loop and surface thermal mass. The mean delivered power exposes the operating regime: it is of order 10^3 W for the residential cases (envelope-conduction / slow-hydronic dominated) but two orders of magnitude larger for the commercial case (roughly $28\times$), because that zone is ventilation/AHU-dominated. This ventilation dominance is the physical reason the commercial controller passes the comfort threshold yet overshoots energy by $+35.3\%$; it also makes a conduction-only time constant ill-defined from total power for that case.

Table 5: Testcase architecture and identified physics. C_{zon} in 10^5 J/K ; equivalent mass = $C_{\text{zon}}/c_{p,\text{air}}$; mean delivered power from the Stage-C telemetry.

Testcase	Heat source	C_{zon}	eq. mass (kg)	mean power (W)	Thermal regime
bestest_air (source)	air supply	4.413	439	—	reference
hydronic heat pump	heat pump	8.348	831	2,778	conduction-dominated
hydronic (boiler)	boiler/radiator	8.623	858	679	slow hydronic
commercial hydronic	district + AHU	8.426	838	76,860	ventilation/AHU-dominated

6.5 Engineering metrics and pass/fail rules

The controller-side pass threshold is normalized by each testcase’s built-in PI baseline, and energy is reported on a separate axis:

$$\tau_k = 1.25 m_{s,\text{PI},k}, \quad \text{PASS}_k^{\text{ctrl}} = \mathbf{1}[m_{s,\text{RL},k} \leq \tau_k], \quad \Delta E_k = 100 \frac{E_{\text{RL},k} - E_{\text{PI},k}}{E_{\text{PI},k}}. \quad (3)$$

The threshold value 1.25 was pre-registered, not chosen post hoc. Surrogate-side transfer is measured by the relative rollout-RMSE improvement after full Stage A/B/C recalibration, and the physical transfer diagnostic is the re-identified capacitance ratio:

$$G_k^{\text{RMSE}} = 100 \frac{\text{RMSE}_{T,k}^{\text{raw}} - \text{RMSE}_{T,k}^{\text{full}}}{\text{RMSE}_{T,k}^{\text{raw}}}, \quad \rho_{C,k} = C_{\text{zon}}^{(k)} / C_{\text{zon}}^{\text{air}}, \quad C_{\text{zon}}^{\text{air}} = 4.413 \times 10^5 \text{ J/K}. \quad (4)$$

Two-axis reporting (m_s threshold and ΔE) is necessary because the commercial testcase passes the m_s bound while consuming substantially more energy than PI.

6.6 Headline transfer matrix

Table 6 is the compact evidence summary and contains two distinct stories. The surrogate component transfers strongly: full Stage A/B/C improves RMSE by 60.2–87.8% on all three target testcases. The frozen controller does not transfer uniformly: the two residential hydronic cases fail the pre-registered $1.25\times$ PI threshold, while the commercial stretch case passes the threshold but pays a +35.3% energy penalty.

Table 6: Block 3 transfer matrix. Threshold = $1.25 m_{s,PI}$ per testcase; data sources in roadmap Section 15. [†] threshold PASS, not deployment-ready (see ΔE).

Testcase	m_s^{RL}	m_s^{PI}	τ_k	Ctrl.	$\Delta E\%$	Raw RMSE	Full RMSE	Gain%	ρ_C
hydronic heat pump	0.665	0.464	0.579	FAIL	-7.3	1.421	0.565	60.2	1.892
hydronic (boiler)	0.976	0.750	0.938	FAIL	-5.8	2.666	0.335	87.4	1.954
commercial hydronic	0.431	0.628	0.785	PASS [†]	35.3	1.952	0.238	87.8	1.909

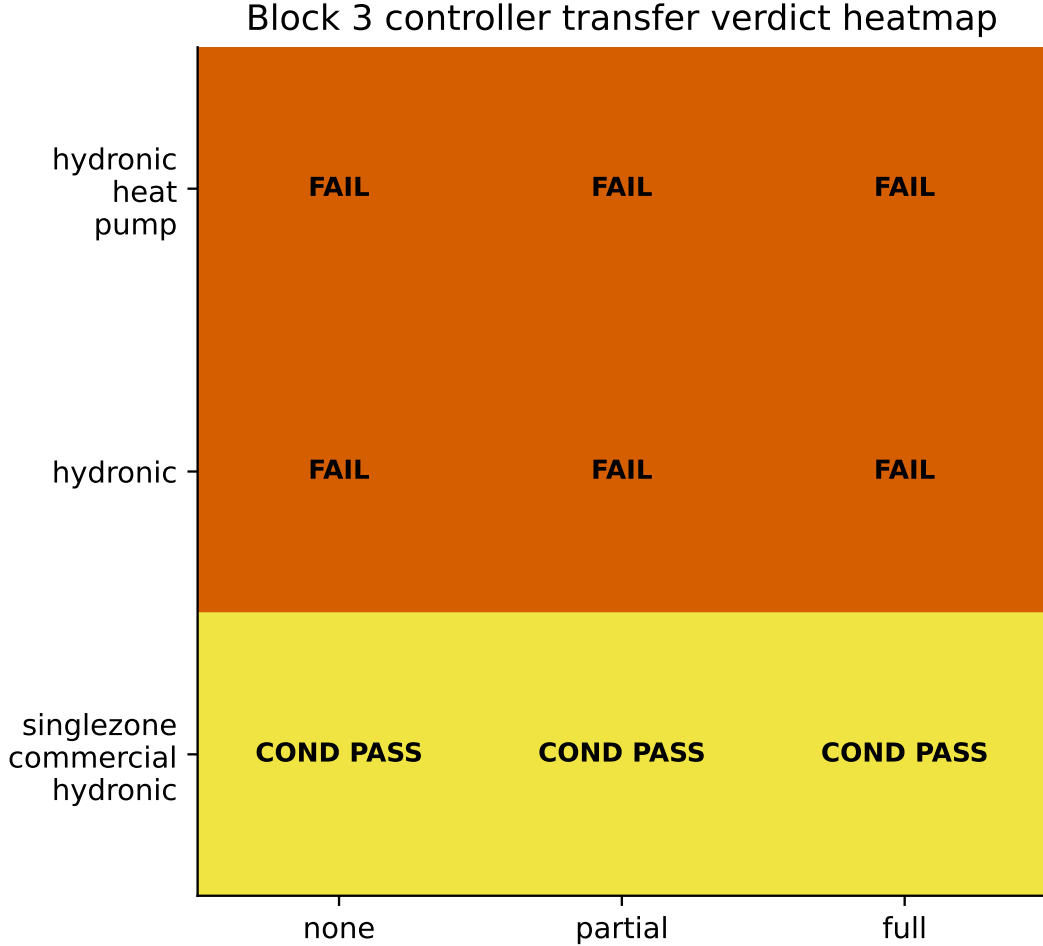


Figure 4: Controller-transfer verdict heatmap. Residential hydronic cases fail the controller threshold; the commercial case passes the threshold but carries a separate energy caveat.

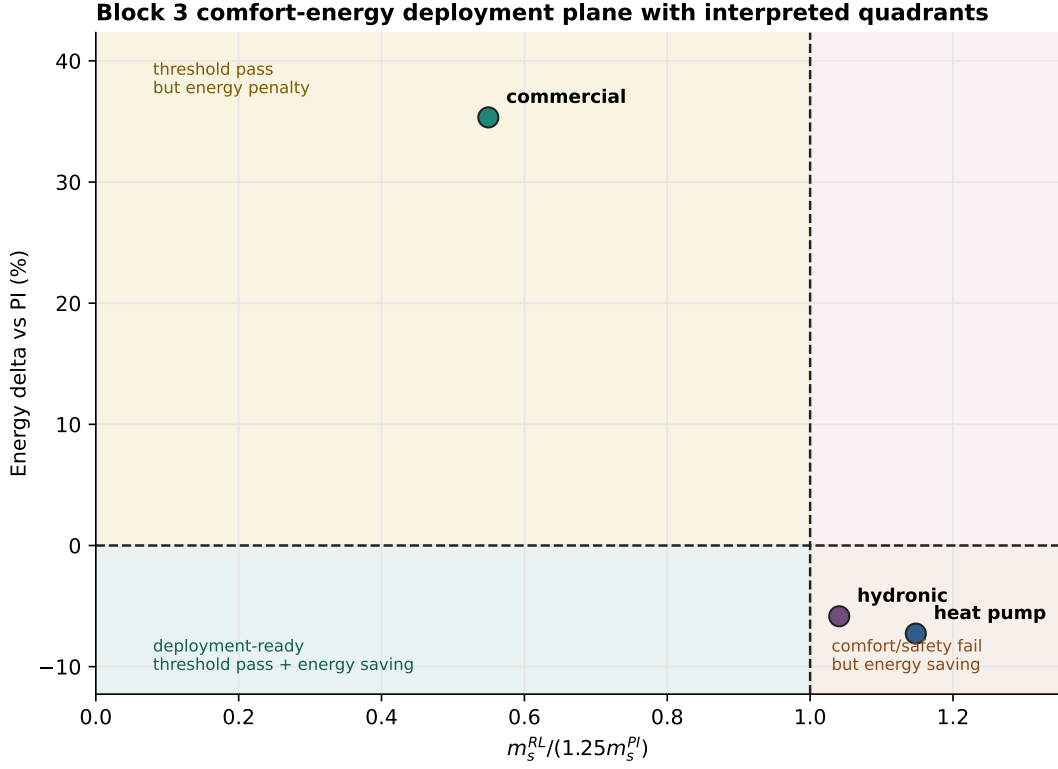


Figure 5: Comfort–energy deployment plane. The residential cases save energy but fail the comfort/safety threshold; the commercial case passes the threshold but moves into the energy-penalty quadrant.

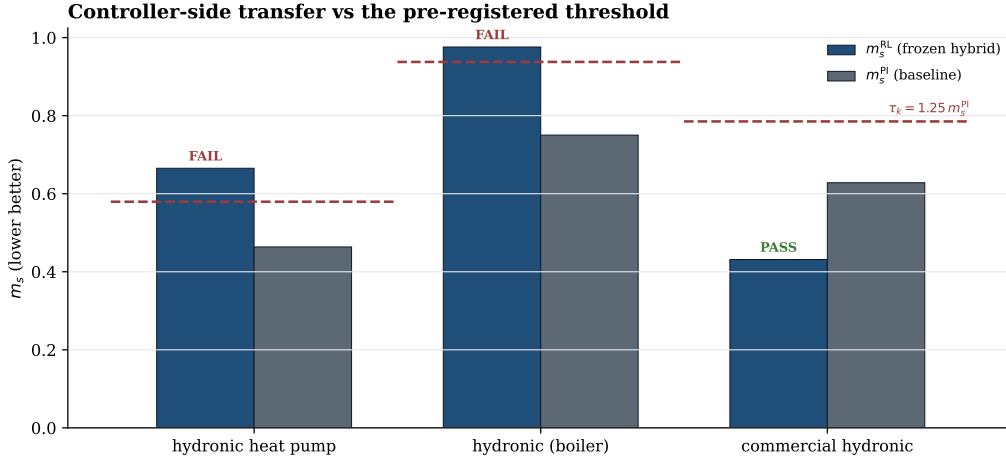


Figure 6: Controller-side transfer: frozen-hybrid m_s^{RL} against the PI baseline m_s^{PI} and the pre-registered threshold $\tau_k = 1.25 m_s^{PI}$ (dashed). The two residential cases exceed the threshold (FAIL); the commercial case is below it (PASS, with the separate energy caveat).

6.7 Primary testcase: `bestest_hydraulic_heat_pump`

The primary testcase was pre-registered as the easiest target (closest to `bestest_air` in envelope class, but with a hydraulic heat-pump actuator). The yearly PI baseline is $m_s = 0.464$ with threshold $\tau = 0.579$. The frozen hybrid controller obtains $m_s = 0.665$ with energy -7.3% relative to PI, so the controller verdict is FAIL: the policy saves energy but violates comfort too often. Full Stage A/B/C recalibration instead succeeds (Table 7): $RMSE_T = 0.565^\circ\text{C}$ and $C_{zon} = 8.348 \times 10^5 \text{ J/K} = 1.892 \times \text{bestest_air}$. The component-level split is explicit: surrogate recalibration transfers; the frozen controller does not.

Table 7: Primary-testcase per-regime detail. m_s is invariant across rows because the controller is frozen by manifest scope; only surrogate fidelity changes (data: roadmap Section 15).

Regime / artifact	RMSE _T (°C)	Power MAE (W)	m_s^{RL}	Status
PI baseline (yearly)	—	—	0.464	reference
none: frozen controller	—	—	0.665	control FAIL
partial: Stage C top-5%	1.006	2336	0.665	diagnostic
partial: all-rows power head	0.977	1765	0.665	diagnostic (power)
partial: all-rows heads	0.781	1768	0.665	surrogate cond. pass
full: Stage A/B/C	0.565	1767	0.665	surrogate PASS, ctrl FAIL

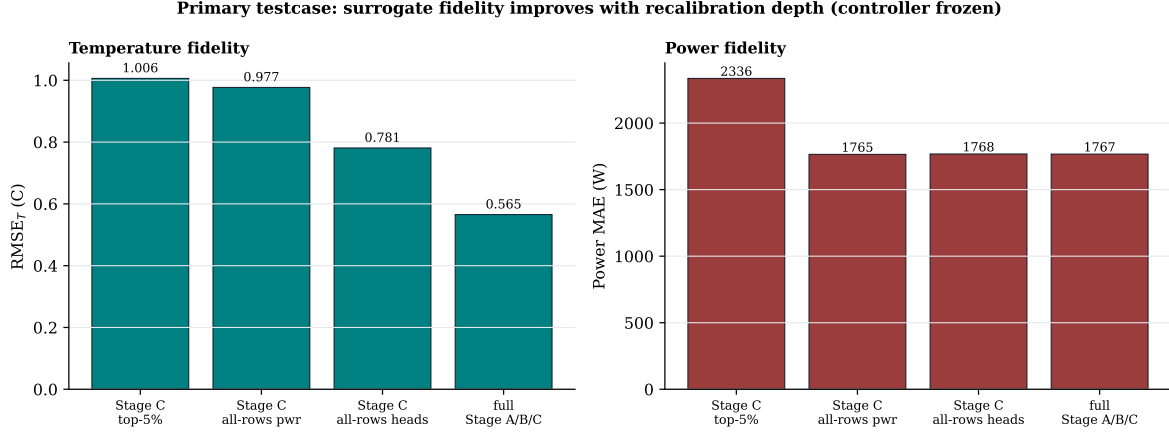


Figure 7: Primary-testcase surrogate-fidelity progression across recalibration regimes. Temperature and power fidelity improve monotonically toward full Stage A/B/C, while the frozen controller’s live m_s is invariant by manifest scope.

6.8 Secondary testcase: `bestest_hydronic`

The secondary testcase tests whether the primary failure was specific to heat-pump nonlinearities. It was not. The residential boiler/radiator case has PI $m_s = 0.750$ and threshold $\tau = 0.938$; the frozen hybrid controller obtains $m_s = 0.976$ with energy -5.8% relative to PI — replicating the residential controller-side failure (energy saved, comfort threshold not met). Full Stage A/B/C again succeeds: RMSE_T drops from 2.666 to 0.335 °C (a 87.4% gain) and $C_{\text{zon}} = 1.954 \times \text{bestest_air}$. The N=2 residential pattern is therefore consistent: frozen-controller transfer fails; surrogate recalibration transfers.

6.9 Stretch testcase: `singlezone_commercial_hydronic`

The stretch testcase was pre-registered as the hardest, most informative falsification probe. The manifest predicted that mode=none controller transfer would FAIL (a-priori probability 0.80) and that the commercial-scale C_{zon} would likely be scale-dependent (hypothesis B, $[3, 10] \times$, a-priori 0.50). The observed result is mixed and scientifically useful. The frozen controller obtains $m_s = 0.431$ against PI $m_s = 0.628$ and threshold 0.785, so it *passes* the m_s criterion — but it consumes 35.3% more energy than PI, so the correct reading is threshold PASS, not deployment-ready PASS. The surrogate side is again strong: full Stage A/B/C reduces RMSE_T from 1.952 to 0.238 °C (87.8%), and the identified C_{zon} ratio is $1.909 \times$ — inside the lower-probability uniform hypothesis A, not the predicted $3\text{--}10 \times$ range.

6.10 Surrogate-side transfer and C_{zon} consistency

Full Stage A/B/C improves target rollout RMSE on all three testcases (gains 60.2%, 87.4%, 87.8%), supporting the weak surrogate-side transfer hypothesis on $N = 3$ (Figure 8). The strongest physical

finding is the consistency of the re-identified capacitance: the three hydronic testcases cluster tightly around $1.918\times$ the `bestest_air` value (Table 8),

$$\rho_C = \{1.892, 1.954, 1.909\}, \quad \bar{\rho}_C = 1.918, \quad s_{\rho_C} = 0.032. \quad (5)$$

The calibrated thermal capacitance is thus more closely associated with hydronic-family physics than with the obvious commercial zone-size proxy: despite the stretch testcase having an order-of-magnitude larger zone volume, its ratio sits within $\pm 2\%$ of the family mean. This generalizes the calibration component of the v3.5 surrogate beyond a single building. The uniformity claim is, however, established on $N = 3$ single-zone hydronic cases and is therefore a hydronic-family observation rather than a statistically powered law; it is not asserted for radiant, VRF, multi-zone, or cross-climate cases (Section 6.13).

Table 8: C_{zon} re-identification across the `bestest_air` baseline and the three hydronic testcases (data: roadmap Section 15).

Testcase	C_{zon} (J/K)	ρ_C vs <code>bestest_air</code>
<code>bestest_air</code> (Block 1)	4.413×10^5	1.000 (baseline)
hydronic heat pump	8.348×10^5	1.892
hydronic (boiler)	8.623×10^5	1.954
commercial hydronic	8.426×10^5	1.909
hydronic-family mean $\pm$ std	—	1.918 ± 0.032

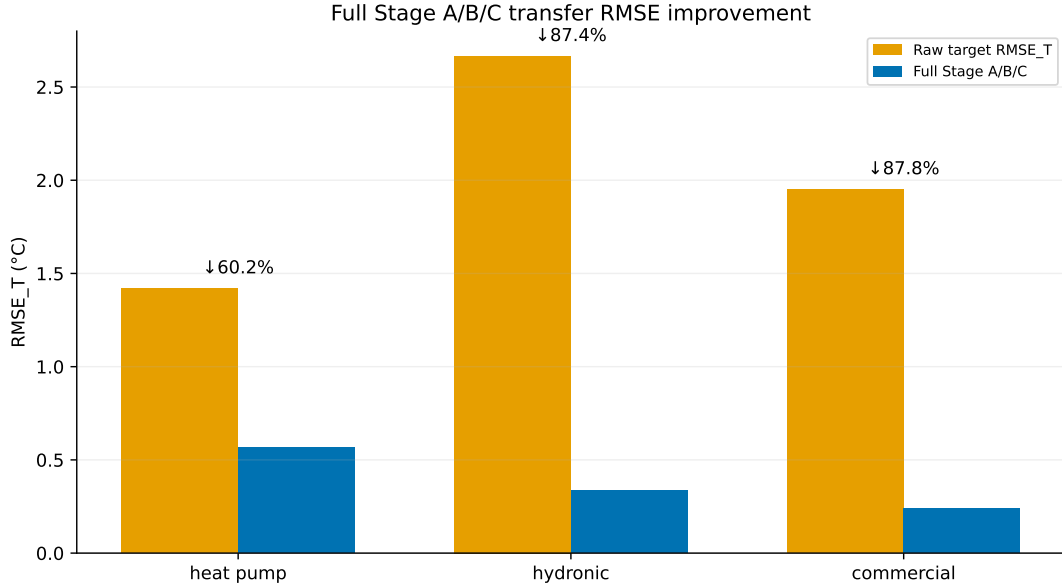


Figure 8: Full Stage A/B/C transfer RMSE improvement on the three hydronic-family target testcases.

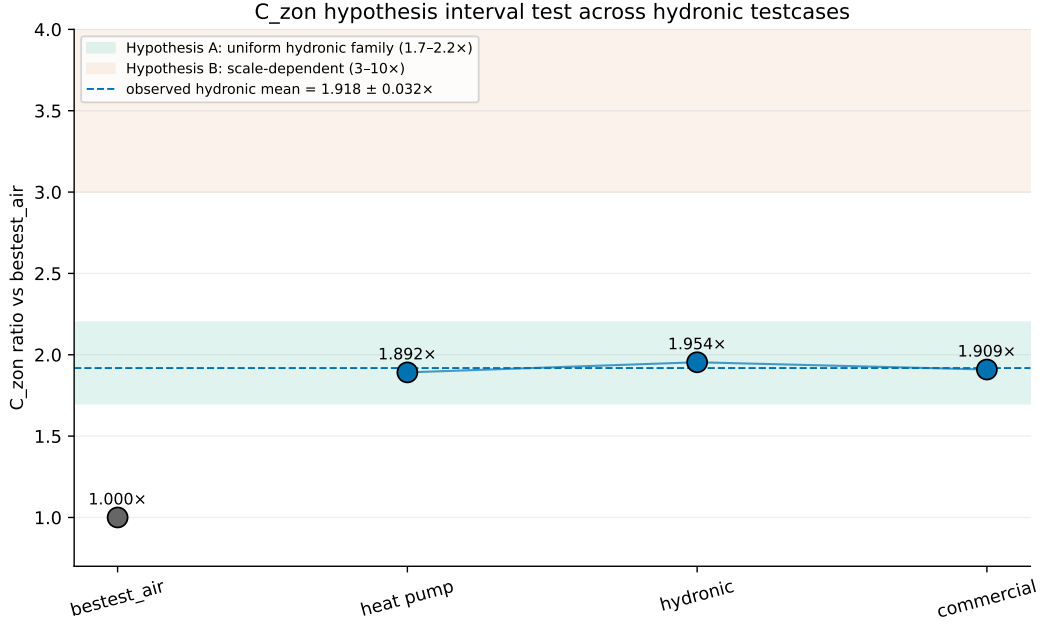


Figure 9: C_{zon} hypothesis-interval diagnostic. The observed ratios fall inside the uniform hydronic-family hypothesis A ($1.7\text{--}2.2\times$) and outside the scale-dependent hypothesis B ($3\text{--}10\times$).

6.11 Hypothesis closure

Table 9 closes the pre-registered hypotheses. The methodological point is that Block 3 does not collapse the evidence into a single label: surrogate transfer and controller transfer diverge.

Table 9: Pre-registered hypothesis closure (manifest `hypothesis_status_final`, audit anchor 7ada793).

Hypothesis	Claim	Verdict	Evidence
H1 (strong)	Frozen mode=none recipe gives deployment-ready transfer.	FALSIFIED	Residential cases fail the $1.25\times$ PI threshold; commercial is threshold-pass only with $+35.3\%$ energy.
H2 (medium)	Partial Stage C recovers controller transfer.	FALSIFIED (structural)	Partial recalibrates only the surrogate; the controller is frozen by scope, so live KPI cannot change.
H3 (surrogate)	Full Stage A/B/C gives a usable target surrogate.	SUPPORTED (N=3)	RMSE _T gains 60.2%, 87.4%, 87.8%; C_{zon} re-identified consistently.
H3 (controller)	Frozen controller transfers given a valid adapter.	FALSIFIED	Residential cases fail comfort; commercial passes safety but overuses energy — regime-dependent failure.
hierarchy	Verdict is monotone in recalibration depth.	SUPPORTED	Controller KPI invariant under surrogate-only recalibration; surrogate fidelity improves with full recalibration.

Hypothesis closure and pre-registered predictions vs observed outcomes		
Hypothesis / prediction	Observed closure	Evidence type
● H1 strong	FALSIFIED	frozen recipe direct transfer
● H2 medium	FALSIFIED BY STRUCTURAL DEFINITION	partial recalibration
● H3 surrogate	SUPPORTED ON N3	full Stage A/B/C RMSE
● H3 controller	FALSIFIED WITH REGIME DEPENDENT FAILURE MODES	controller full regime
● Stretch none	expected FAIL → observed PASS	commercial mode=none
● Stretch C _{zon}	expected scale-dependent → observed 1.909×	commercial C _{zon}
● Stretch RMSE	expected 50-90% → observed 87.81%	commercial full recalibration

Figure 10: Hypothesis closure matrix: Block 3 supports surrogate-side transferability but falsifies deployment-ready frozen-controller transfer.

6.12 Pre-registered predictions versus outcomes

The stretch testcase carried numerical a-priori predictions (audit anchor 645626e, logged before any commercial-hydronic run). Reporting the predicted-vs-observed mapping is the central Popperian discipline of Block 3 (Table 10): the evidence moved the interpretation toward lower-prior hypotheses rather than confirming only what was expected. Two predictions were falsified (the mode=none controller FAIL, and the scale-dependent C_{zon} hypothesis) and three were confirmed.

Table 10: Stretch-testcase pre-registered predictions versus observed outcomes (manifest stretch_testcase_predictions).

Prediction	Pre-reg.	A-priori	Observed	Verdict
mode=none controller verdict	FAIL	0.80	threshold PASS	FALSIFIED
full surrogate RMSE gain	[50, 90]%	0.70	87.8%	CONFIRMED
C_{zon} hyp. A (uniform)	[1.7, 2.2]×	0.35	1.91×	CONFIRMED
C_{zon} hyp. B (scale-dep.)	[3, 10]×	0.50	1.91×	FALSIFIED
C_{zon} hyp. C (failure)	non-convergence	0.15	converged (N=3)	FALSIFIED

6.13 Limitations

- **Data provenance.** The live KPIs (m_s , $RMSE_T$, energy, RMSE gains, and the C_{zon} ratios) are read directly from the `reports/block3_*` artifacts; the pre-registered hypotheses, a-priori probabilities, predictions, and actuator-adapter mappings are verified literals from the audit-frozen manifest and adapter configs; and absolute C_{zon} values are reconstructed as $\rho_{C,k} \times 4.413 \times 10^5$ J/K from the data-driven ratio.
- **Single run per cell.** Each testcase×regime cell is a single live BOPTEST run (the manifest caps $N = 3$ per cell and excludes a seed cascade), so the transfer KPIs carry no seed-variance interval; the $\bar{\rho}_C$ spread is across testcases, not seeds.
- **Frozen-controller scope.** By manifest design the controller is never fine-tuned on the target testcase; Block 3 therefore measures the transferability of the *calibration pipeline*, not of an adapted controller.
- **N=3 hydronic family.** The uniform- C_{zon} finding (1.918 ± 0.032) is established on three single-zone hydronic cases; it is not claimed for radiant, VRF, multi-zone, or cross-climate cases.
- **Single weather source.** All testcases use one weather-file source, so cross-climate generalization is explicitly out of scope.

- **Adapter-mediated transfer.** Controller transfer is mediated by pre-registered rule-based actuator adapters; a different adapter design could change the controller-side verdict.

6.14 Results III conclusion

Block 3 establishes a precise component-level transferability boundary. The inverse-calibration component of the v3.5 surrogate transfers strongly to the hydronic family: full Stage A/B/C improves RMSE by 60–88% and re-identifies C_{zon} at a consistent $1.918 \pm 0.032 \times$ the `bestest_air` value, independent of the obvious zone-size proxy. The frozen-controller component does *not* transfer as a deployment-ready policy: the two residential hydronic cases save energy but fail the comfort/safety threshold, and the commercial stretch case passes the m_s threshold only at a +35.3% energy cost. The transferability boundary therefore lies at the controller–adapter interface, not in the surrogate’s ability to recalibrate target physics.

The pre-registration discipline is the methodological contribution: hypotheses, predictions, and the $1.25 \times$ threshold were frozen before the runs, so the two falsifications (mode=none controller FAIL on the stretch case; scale-dependent C_{zon}) are genuine Popperian updates rather than post-hoc narrative. The precise next step for Block 4 follows directly: the target surrogate is accurate enough to be a candidate training environment, so target-specific controller fine-tuning on the fully recalibrated surrogate — not a frozen source controller — is the falsifiable path to deployment-ready hydronic transfer.